SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BCSE302P - DATABASE SYSTEMS LAB

Semester : Winter Semester 25-26 **Faculty : Dr.S.Geetha**

EXERCISE 6

Title: Sub Query

1. OBJECTIVE

- To understand the concept of subqueries in SQL
- To apply single-row and multi-row subqueries
- To use subqueries with SELECT, WHERE, and HAVING clauses
- To perform nested queries for advanced data retrieval

2. SUBQUERIES

- A subquery is a SQL query nested inside another query.
- The result of the inner query is used by the outer query.
- In other words we can say that a Subquery is a query that is embedded in WHERE/HAVING/FROM clause of another SQL query.

Usage:

- Subqueries can be used with SELECT, UPDATE, INSERT, DELETE statements along with expression operator. It could be equality operator or comparison operator such as =, >, =, <= and Like operator.
- A subquery is a query within another query. The outer query is called as main query and inner query is called as subquery.
- The subquery generally executes first, and its output is used to complete the query condition for the main or outer query.
- Subquery must be enclosed in parentheses.
- Subqueries are on the right side of the comparison operator.
- ORDER BY command cannot be used in a subquery.
- GROUPBY command can be used to perform same function as ORDER BY command.
- Use single-row operators with singlerow Subqueries. Use multiple-row operators with multiple-row Subqueries.

Types:

- Single-row subqueries
- Multi-row subqueries
- Multiple-column subqueries
- Correlated subqueries

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3. STRING FUNCTIONS

SELECT column_name
FROM table_name
WHERE column_name operator (SELECT column_name FROM table_name);

STUDENT TABLE

student_id	student_name	dept	age
101	Arun	CSE	20
102	Banu	ECE	21
103	Charan	CSE	19
104	Divya	MECH	22
105	Eshwar	CSE	21
106	Farah	CIVIL	20

EMPLOYEE TABLE

emp_id	emp_name	dept	salary	age	join_date
1	Anand	IT	45000.75	28	2021-06-15
2	Bhavya	HR	38000.40	32	2020-03-10
3	Charan	IT	52000.60	26	2022-01-20
4	Divya	FINANCE	60000.00	35	2019-08-05
5	Ezhil	HR	40000.20	29	2021-11-12
6	Farooq	IT	30000.90	24	2023-02-01

COURSE TABLE

course_id	course_name	credits
201	DBMS	4
202	OS	3
203	CN	3
204	AI	4
205	ML	4

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ENROLLMENT TABLE

enroll_id	student_id	course_id	marks	semester
1	101	201	78	3
2	101	204	85	3
3	102	201	68	3
4	102	203	72	3
5	103	204	90	2
6	103	202	75	2
7	104	205	60	4
8	105	201	88	3
9	105	203	82	3
10	106	205	55	4

A. SINGLE-ROW SUBQUERY

Meaning

- A single-row subquery returns **exactly one row and one column** as its result.
- It is used when the outer query needs **one specific value** for comparison.

Execution Method

• The inner (sub)query is executed **first**
• It returns a **single value**
• The outer query then uses this value in its condition

Operators Used

= , > , < , >= , <= , <>

Template Syntax

```
SELECT column_name
FROM table_name
WHERE column_name operator (
   SELECT column_name
   FROM table_name
);
```

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1. Display employees earning more than the average salary.
```
SELECT emp_name, salary
FROM EMPLOYEE
WHERE salary > (SELECT AVG(salary) FROM EMPLOYEE);
```

B. MULTI-ROW SUBQUERY

Meaning

- A multi-row subquery returns multiple rows of values. It is used when the condition needs to compare one value with a set of values.

Execution Method

- The subquery executes first
- It returns **multiple rows**
- The outer query compares its values using special operators

Operators Used

- IN , ANY , ALL

Template Syntax

```
SELECT column_name
FROM table_name
WHERE column_name operator (
   SELECT column_name
   FROM table_name
);
```

2. Display students who scored marks greater than any student in semester 4.
```
SELECT student_id, marks
FROM ENROLLMENT
WHERE marks > ANY (
   SELECT marks FROM ENROLLMENT WHERE semester = 4);
```

C. SUBQUERY USING IN

Meaning

- A multiple-column subquery returns **more than one column**.
- It is used when comparison requires **a combination of columns**.

Execution Method

- The subquery executes first
- It returns multiple columns and rows
- The outer query compares column combinations using tuples

Operators Used

IN , =

Template Syntax

```
SELECT column1, column2
FROM table_name
WHERE (column1, column2) IN (
   SELECT column1, column2
   FROM table_name
);
```

3. Display students enrolled in course 201.

```
SELECT student_name
FROM STUDENT
WHERE student_id IN (
   SELECT student_id FROM ENROLLMENT WHERE course_id
= 201 );
```

D. CORRELATED SUBQUERY

Meaning

- A correlated subquery depends on the **outer query for its values**.
- It is executed **once for each row** processed by the outer query.

Execution Method

- Outer query selects one row
- Subquery executes using that row's value
- Process repeats for every row in the outer query

Key Feature

- Subquery **cannot run independently**

Template Syntax

```
SELECT column_name
FROM table_name alias1
WHERE column_name operator (
   SELECT column_name
   FROM table_name alias2
   WHERE alias2.column = alias1.column
);
```

4. Display employees who earn the highest salary in their department.

```
SELECT emp_name, dept, salary
FROM EMPLOYEE e
WHERE salary = (
   SELECT MAX(salary)
   FROM EMPLOYEE
   WHERE dept = e.dept );
```

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4. EXPECTED OUTCOMES

After executing this experiment, the student will be able to:

- Write and execute nested SQL queries
- Differentiate between single-row and multi-row subqueries
- Apply subqueries for complex data analysis
- Use correlated subqueries effectively

5. EXPERIMENT QUESTIONS

What is a subquery?

Difference between subquery and join?

What is a correlated subquery?

Difference between ANY and ALL?

Can a subquery return multiple columns?

PRACTICE QUESTIONS

A. SINGLE-ROW SUBQUERY

1. Display the details of employees whose salary exceeds the salary received by the most recently joined employee.
2. List employees whose age is higher than that of the youngest employee in the organization.
3. Retrieve employees earning more than the salary drawn by the employee named *Bhavya*.
4. Display students who scored more marks than the marks obtained by student *Charan* in any one of his enrolled courses.
5. Find employees whose salary is higher than the salary of the employee who joined first in the organization.

B. MULTI-ROW SUBQUERY

1. Display students who have scored more marks than students belonging to semester 4.

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2. List employees whose salary is lower than the salaries drawn by employees working in the IT department.
3. Retrieve students who have enrolled in courses already registered by student *Arun*.
4. Display employees working in departments where at least one employee has a salary above 55,000.
5. List students whose marks are higher than the marks obtained in course *ML*.

C. MULTIPLE-COLUMN SUBQUERY

1. Display students who are enrolled in the same course and semester as student *Arun*.
2. Retrieve employees who share the same department and salary combination as any other employee.
3. Display enrollment details that match the course and semester combinations of CSE students.
4. List employees whose department and age match those of any HR employee.
5. Display students registered in the same course–semester pair as those in semester 3.

D. CORRELATED SUBQUERY

1. Display employees who earn more than all other employees in their respective departments.
2. Retrieve students who secured the highest marks in each course.
3. Display employees whose salary is higher than the salary of colleagues working in the same department.
4. List students whose marks are higher than the marks of other students in the same semester.
5. Display employees who joined earlier than others in their respective departments.